

# City General Hospital

## 30-Day Readmission Risk Assessment

|               |                     |            |                         |
|---------------|---------------------|------------|-------------------------|
| Patient ID:   | 20251103-COP-8360e7 | Generated: | November 03, 2025 02:11 |
| Patient Name: | JOSHUA MO           | Age:       | 32 years                |
| Sex:          | Male                | Disease:   | COPD                    |

30-Day Readmission Risk: **81.3%** | Classification: **HIGH RISK** (Threshold: 50%)

### Index Admission Summary

|                         |     |                        |        |
|-------------------------|-----|------------------------|--------|
| Length of Stay:         | N/A | Discharge Destination: | N/A    |
| Prior Admissions (90d): | N/A | Comorbidities Count:   | N/A    |
| Follow-up Scheduled:    | N/A | Admission Type:        | Urgent |

### Top Contributing Risk Factors (SHAP Analysis)

| Feature                 | Value | Contribution | Impact        | Interpretation                                                                                                                                         |
|-------------------------|-------|--------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| FEV1_percent            | 0.00  | +1.928       | ↑ Higher Risk | FEV1_percent influences readmission risk.. This feature has a major effect and increases the readmission risk.                                         |
| Oxygen_saturation       | 0.00  | +0.992       | ↑ Higher Risk | Low oxygen indicates inadequate ventilation or exacerbation. (0.00% — below ref 92-100%). This feature has a moderate effect and increases the read... |
| Use_of_home_oxygen      | 0.00  | -0.768       | ↓ Lower Risk  | Use_of_home_oxygen influences readmission risk.. This feature has a moderate effect and reduces the readmission risk.                                  |
| age                     | 32.00 | -0.677       | ↓ Lower Risk  | age influences readmission risk.. This feature has a moderate effect and reduces the readmission risk.                                                 |
| length_of_stay          | 0.00  | -0.498       | ↓ Lower Risk  | length_of_stay influences readmission risk.. This feature has a minor effect and reduces the readmission risk.                                         |
| prior_admissions_90d    | 0.00  | -0.461       | ↓ Lower Risk  | prior_admissions_90d influences readmission risk.. This feature has a minor effect and reduces the readmission risk.                                   |
| Prior_exacerbations_1yr | 0.00  | -0.415       | ↓ Lower Risk  | Prior_exacerbations_1yr influences readmission risk.. This feature has a minor effect and reduces the readmission risk.                                |
| followup_scheduled      | 0.00  | +0.367       | ↑ Higher Risk | followup_scheduled influences readmission risk.. This feature has a minor effect and increases the readmission risk.                                   |

### Clinical Summary

This 32 years male patient presents with a **high risk 30-day readmission risk (81.3%)** following discharge for COPD management. The risk assessment threshold for this condition is 50%, placing this patient significantly above the high-risk threshold.

**Primary Risk Drivers:** FEV1\_percent, Oxygen\_saturation, Use\_of\_home\_oxygen are identified as major contributing factors.

**Recommendation:** Close follow-up and aggressive management of identified risk factors is recommended to prevent readmission.

## Clinical Management Recommendations

- **Primary Disease Management:** Review and optimize current treatment plan for COPD. Consider consultation with appropriate specialists.
- **Medication Reconciliation:** Complete medication review at discharge. Ensure patient understands all medications, dosing, and timing.
- **Care Coordination:** Schedule follow-up appointment within 7 days of discharge. Consider home health services for high-risk patients.
- **Patient Education:** Reinforce medication adherence, warning signs requiring immediate attention, and lifestyle modifications.
- **Laboratory Monitoring:** Schedule appropriate lab work based on disease-specific guidelines and medication monitoring requirements.

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## Medication Recommendations

|             |                     |               |           |
|-------------|---------------------|---------------|-----------|
| Patient ID: | 20251103-COP-8360e7 | Patient Name: | JOSHUA MO |
| Age / Sex:  | 32 years / Male     | Disease:      | COPD      |

**Important:** The following medication recommendations are based on current clinical guidelines and the patient's risk profile. All medications must be reviewed, prescribed, and adjusted by the attending physician based on individual patient factors, allergies, drug interactions, and institutional protocols.

### Recommended Medication Protocol:

#### Bronchodilators:

- LABA + LAMA combo (e.g., Tiotropium + Olodaterol inhaler)
- Short-acting beta-agonist (Albuterol) as rescue inhaler
- Theophylline if persistent symptoms

#### Anti Inflammatory:

- Inhaled corticosteroids (e.g., Fluticasone) if frequent exacerbations
- PDE4 inhibitor (Roflumilast) for severe COPD
- Oral corticosteroids (Prednisone 40mg x 5 days) during exacerbations

#### Oxygen Therapy:

- Long-term oxygen if SpO2 <88% or PaO2 <55 mmHg
- Pulmonary rehabilitation program
- Smoking cessation support (Varenicline or Bupropion)

#### Monitoring:

- Spirometry annually, pulse oximetry, vaccination (flu + pneumococcal)

*Note: These are general guidelines. All medications should be prescribed and adjusted by the attending physician based on individual patient factors.*

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## Potential Disease Progression & Related Conditions

|             |                     |               |           |
|-------------|---------------------|---------------|-----------|
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**Potential Disease Progression & Related Conditions:**

**[HIGH RISK] High-Risk Conditions (Requires Active Prevention):**

**1. Respiratory Failure**

- *Risk Factors:* FEV1 <30%, frequent exacerbations, hypoxemia
- *Typical Time Frame:* 2-5 years
- *Prevention Strategy:* Oxygen therapy, pulmonary rehab, medication adherence

**2. Cor Pulmonale (Right Heart Failure)**

- *Risk Factors:* severe COPD, chronic hypoxia, pulmonary hypertension
- *Typical Time Frame:* 5-10 years
- *Prevention Strategy:* Long-term oxygen, diuretics, treat underlying COPD

**3. Lung Cancer**

- *Risk Factors:* smoking history, age >55, COPD severity
- *Typical Time Frame:* 10-20 years
- *Prevention Strategy:* Smoking cessation, annual low-dose CT screening

**[MODERATE RISK] Moderate-Risk Conditions (Monitor Closely):**

**1. Pneumonia (Recurrent)**

- *Risk Factors:* impaired clearance, immunosuppression
- *Prevention:* Vaccination, inhaler technique, smoking cessation

**Personalized Risk Assessment:**

- Current markers within acceptable ranges. Continue monitoring.

**Disclaimer:** This report is generated by a machine learning model for clinical decision support. All recommendations should be reviewed and approved by qualified healthcare professionals. Medication dosages and treatment plans must be individualized based on patient-specific factors, comorbidities, and current clinical guidelines.